

Anisshka Annand Khadtale, Saachi Dighe, Tiffany Ribeiro

NUDGING OUTPUT ONLINE: A BONUS VS. FIXED-WAGE EXPERIMENT

DECEMBER 4, 2025

GOAL AND PURPOSE



OUR FIELD EXPERIMENT

- An audio transcription containing 10 words in 30 seconds
- The worker has 2 minutes to write everything down from memory



To understand how people encode word lists under light time pressure.



PURPOSE

To test the effect of 2 types of worker incentives on Prolific worker's quantity and quality of work

METHOD

NOV 4: DELIBERATION

NOV 18: LAUNCH+ DATA
COMPLETE

DEC 3: REPORT FINAL
DRAFT

NOV 11: DESIGN
READY

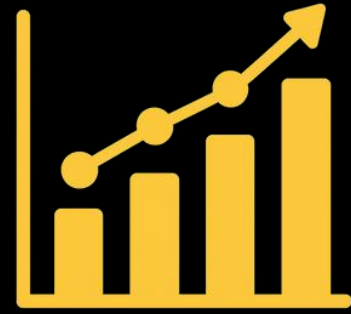
PAYMENT

- Used two different payment methods to determine our main goal
- Base fixed pay - \$2.00
- Base pay \$1.50 with incentives of \$0.10 for every word correct written (maximum 10 words)

QUANTITY/ QUALITY TEST

- Quantity - Word count + Attempt (0/1)
- Quality - Error count + Quality score Subjective

METHOD



QUANTITY

T-Test:

Treatment Group :

Regression **WITH CONTROLLED** Variables:

$$\text{Wordcount} = a + b * \text{treatment}$$

Regression **Without Controlled** Variables :

$$\text{Wordcount} = A + B * \text{Treatment} + C * \text{controlled variables (age, gender, edu)}$$

QUALITY

T-Test:

Treatment Group :

Regression **WITH CONTROLLED** Variables:

$$\text{Quality Score} = a + b * \text{treatment}$$

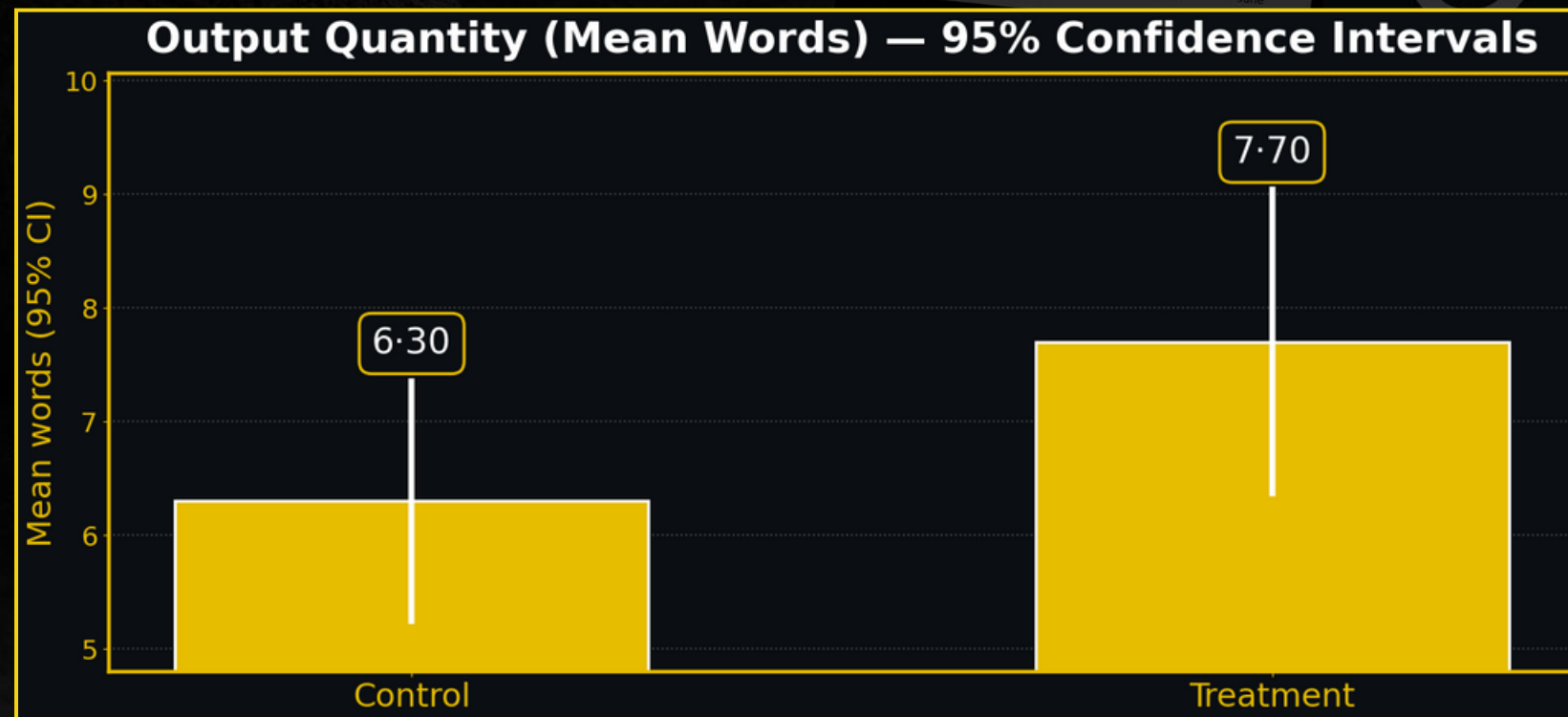
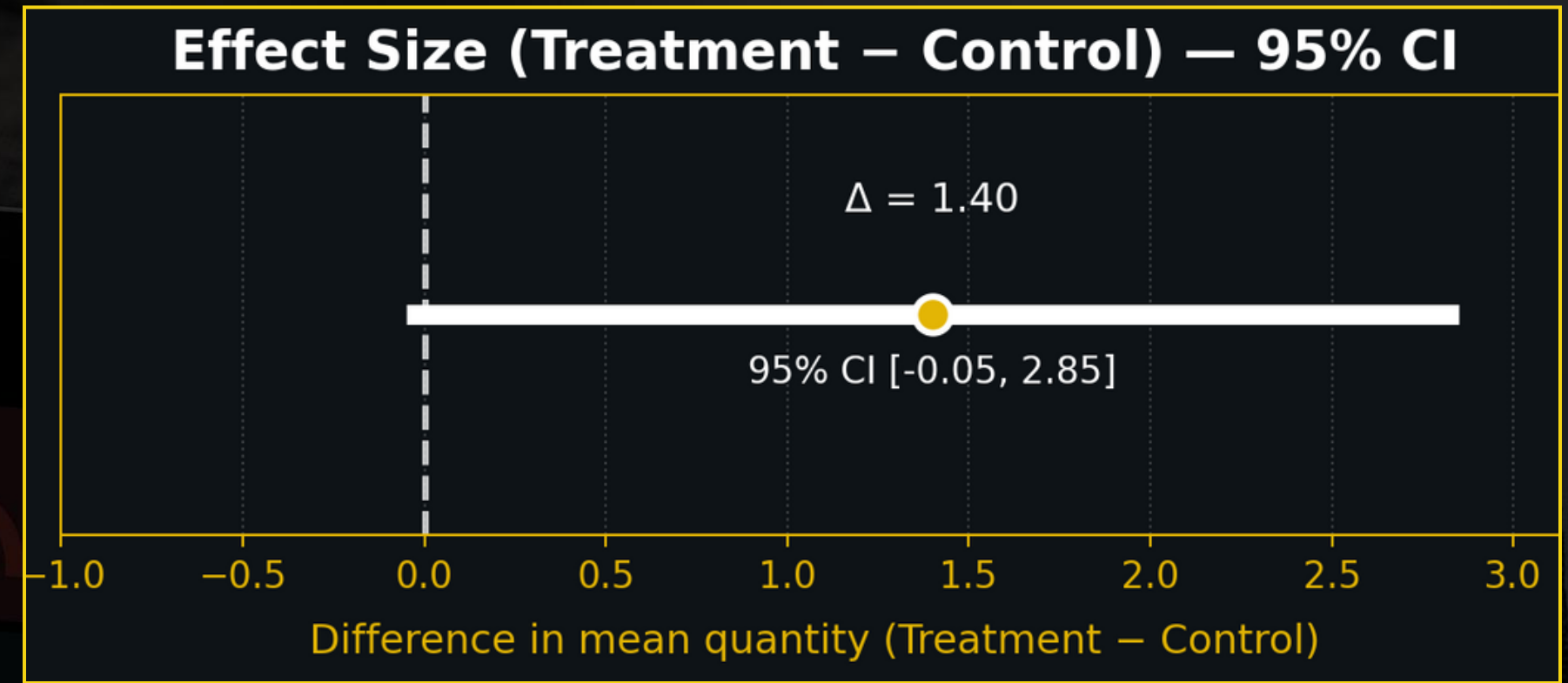
Regression **Without Controlled** Variables :

$$\text{Quality Score} = A + B * \text{Treatment} + C * \text{controlled variables (age, gender, edu)}$$



RESULT 1 : QUANTITY

Effect-size plot: “Estimated gain $\approx +1.4$ words (95% CI $\sim [-0.04, 2.84]$); significant on a one-tailed test (directional hypothesis), marginal on two-tailed.”

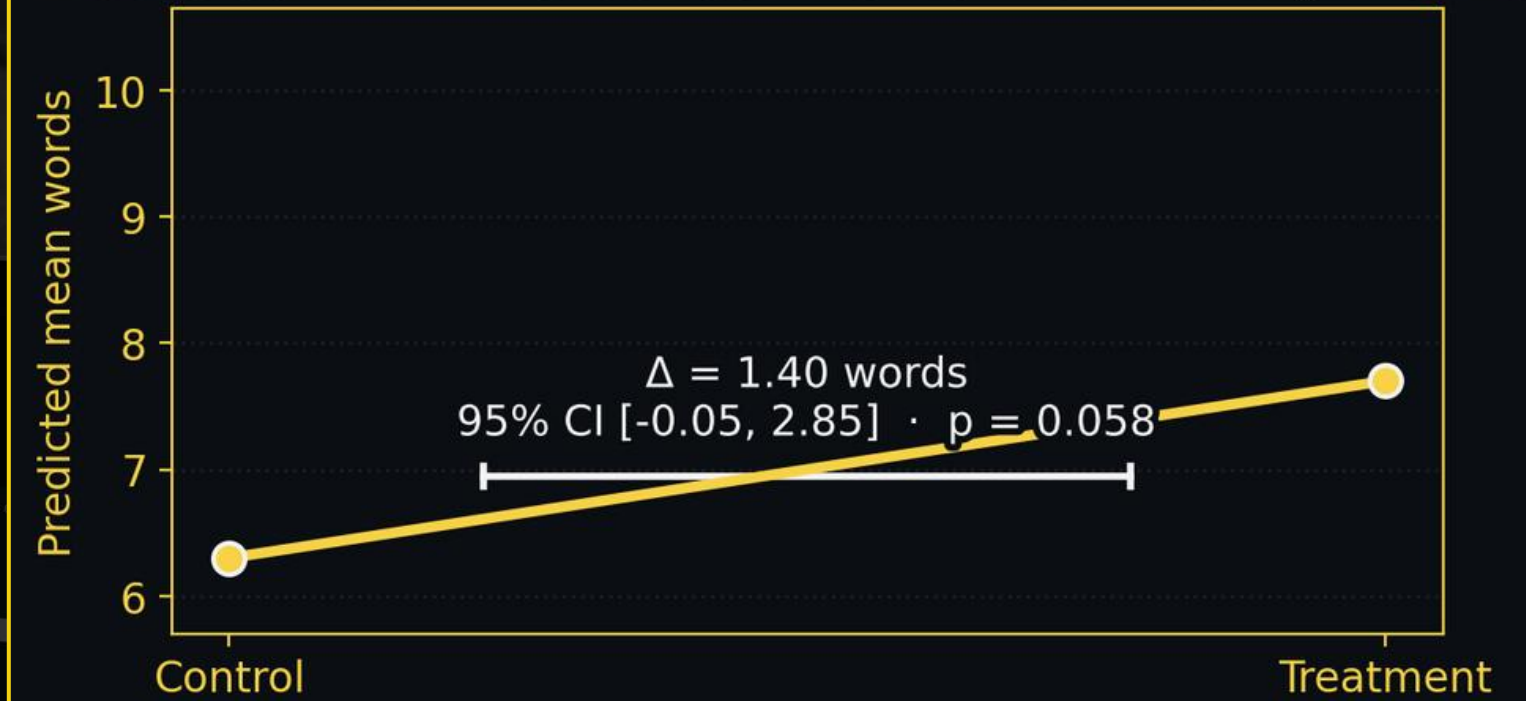


Bar chart: “Treatment workers produced a higher mean wordcount (7.7 vs 6.3); error bars show 95% CIs for each mean.”

RESULT 1: QUANTITY

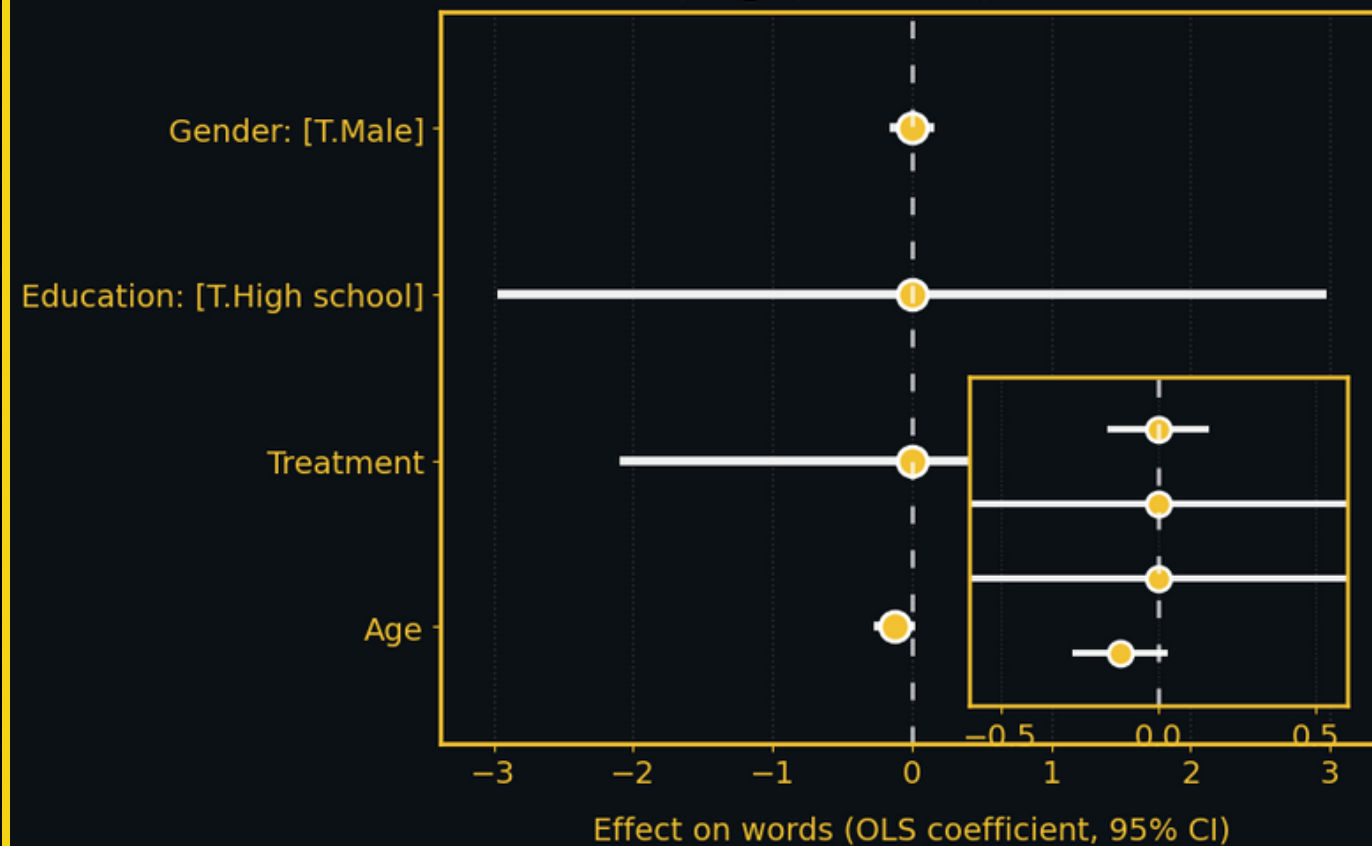
“The treatment increased word count by ~1.40 words; evidence is suggestive but not conclusive ($p=0.058$; 95% CI $[-0.05, 2.85]$).”

Predicted Mean Words: Control vs Treatment (Line Plot)



Bracket represents 95% CI for the difference (T – C)

Effect of Treatment, Age, Gender, Education on Word Count

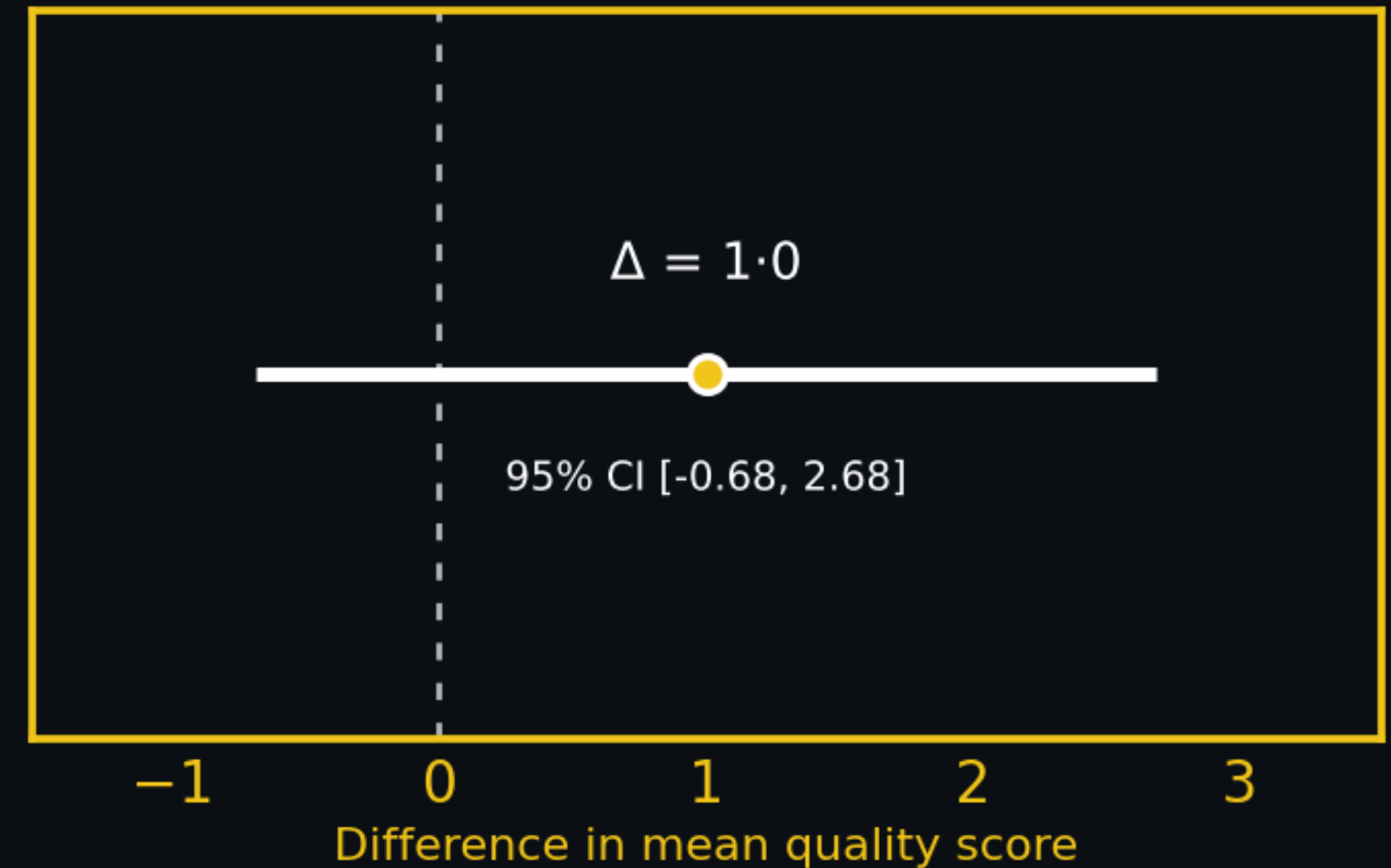


“The coefficient plot shows that treatment, age, gender, and education all have effects centered near zero with 95% CIs crossing zero, meaning none are statistically distinguishable from no effect on word count.”

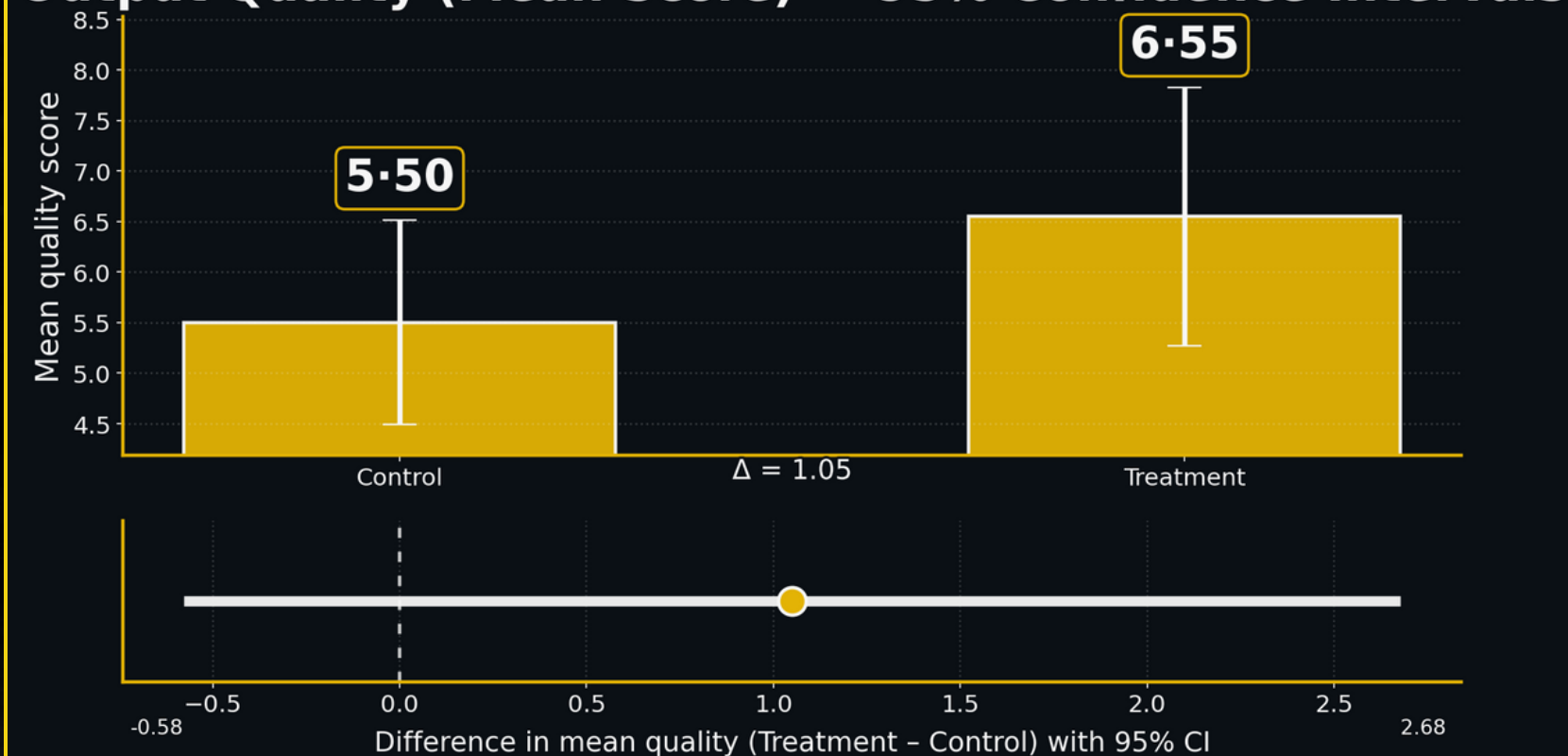
RESULT 2: QUALITY

“**Effect size plot:** The estimated difference is $\Delta \approx +1.05$, with a 95% CI $\approx [-0.63, 2.74]$, which crosses zero—so the increase is not statistically significant at the 5% level.”

Effect Size (Treatment – Control) — 95% CI



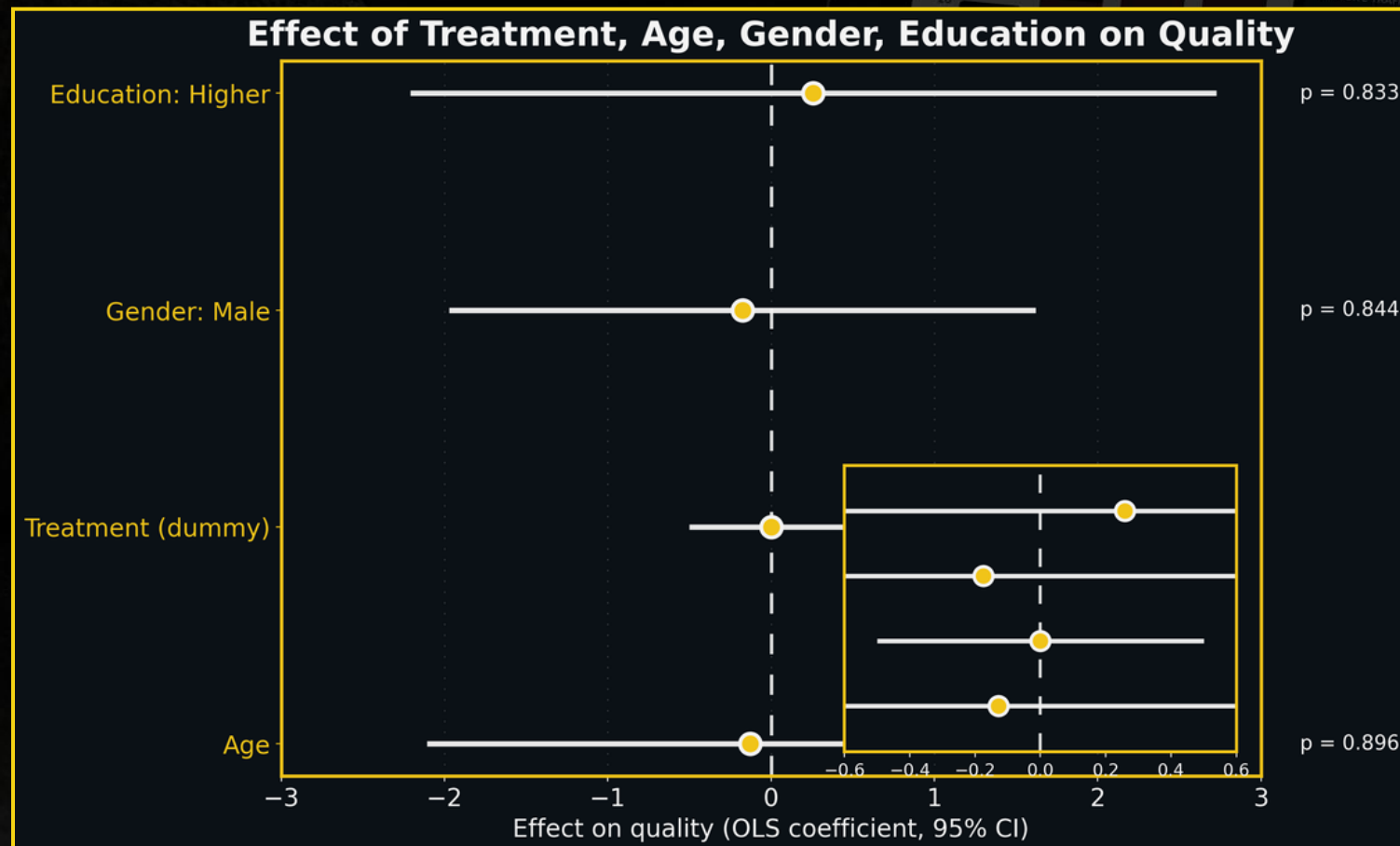
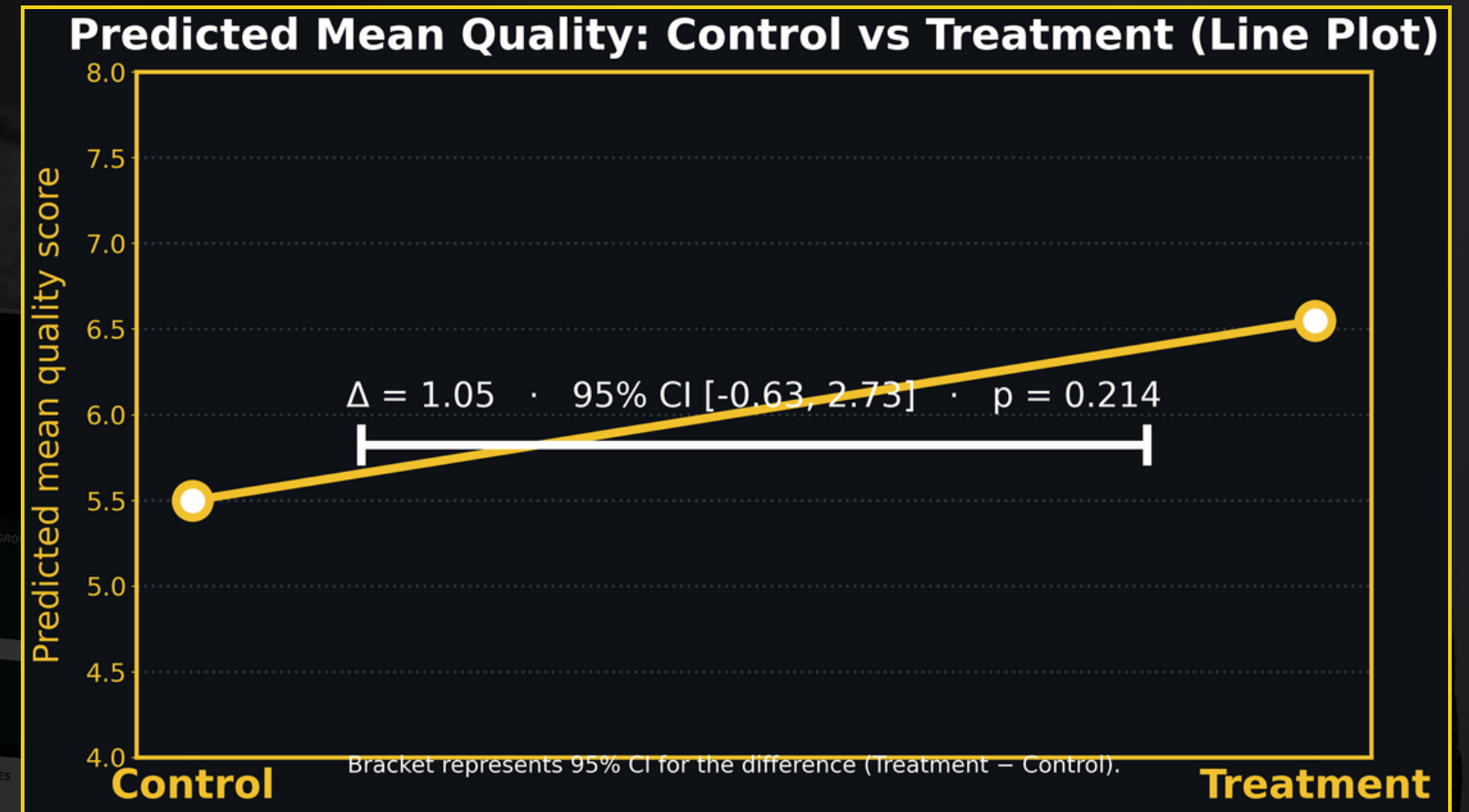
Output Quality (Mean Score) — 95% Confidence Intervals



Bar chart: Treatment’s average quality score (≈ 6.6) is higher than Control’s (≈ 5.5), but the 95% confidence intervals overlap considerably.”

RESULT 2: QUALITY

Higher predicted mean quality score in the treatment group (6.55) than in the control group (5.50), a difference of $\Delta = 1.05$ points. However, the 95% CI for this difference $[-0.63, 2.73]$ includes zero and the p-value = 0.214, so the effect is not statistically significant.



All four OLS coefficients have 95% CIs that cross zero, so none are statistically significant (p 's ≈ 0.83 – 0.90). Point estimates are small (age and male slightly negative, higher education slightly positive), indicating no reliable effect of the treatment or demographics on quality in this sample.

CONCLUSION

- Workers with Contract B (bonus pay) recalled 1-2 more words on average than those with Contract A (\$2 flat).
- Both t-test and regression confirmed the increase in word recall was consistent, not due to chance.
- While Contract B workers recalled more words, their accuracy remained unchanged, with no quality score differences.
- Results were consistent after controlling for age, gender, and education, indicating the contract caused the difference, not the workers themselves.

CONTRACT A

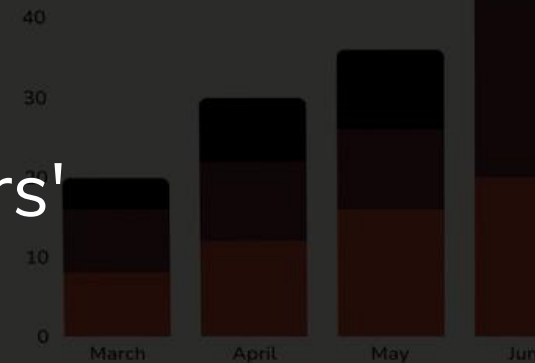
CONTRACT B

DISCUSSION/CAVEATS

- The small sample size ($n=40$) limits the precision of outcome detection, particularly quality.
- Online study conditions varied by workers' environments and attention levels.

FINANCIAL OUTLOOK

INCOME OVERVIEW



Future Research Roadmap

- Longer tasks / more time
- Multiple recall rounds
- Richer / complex scoring

THANK YOU
FOR YOUR ATTENTION
QUESTIONS?

GROUND

HE INDUSTRY'S HISTORY